CATERPILLAR®

DEALER BEST PRACTICE SERIES

Bulk Fuel Filtration

Application **Maintenance** Site Management Component Rebuild Safety MARC Management

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1.0 Introduction

Modern fuel systems use electronic unit injectors which deliver precise amounts of fuel at pressures up to 25,000 psi, and control injection timing to within thousandths of a second. Electronic unit injectors control the performance and fuel economy of the engine and are expensive to replace when worn. A rough estimate of injector replacement cost with parts and labor is approximately $1,000 per cylinder. And, of course, there's the cost of taking a machine out of production.

Injectors operated on clean fuel should last through engine overhaul. The fuel filters on the machine are designed to provide final filtration for moderately clean supply fuel. Machine filtration is not intended to clean fuel contaminated with large amounts of dirt and water. If contaminated fuel is used, the capability of the onboard filtration is overwhelmed and injectors either wear out prematurely or seize.

For a variety of unavoidable reasons, fuel delivered to mine sites is usually contaminated with dirt and water. Because fuel is a very low margin commodity, suppliers almost never provide adequate bulk fuel filtration or exercise recommended storage practices. As a result, customers receive and use contaminated fuel, resulting in premature injector failure and wear out. This causes excessive fuel consumption and often results in mid-life injector set replacement.

Bulk fuel filtration has been used in the aviation industry for more than 50 years to address the same problems. Caterpillar has now adopted this proven technology to help mining customers clean contaminated fuel.

2.0 Best Practice Description

Bulk fuel filtration consists of high capacity filters, which remove both excess dirt and water from the supply fuel before it is put into the machine.

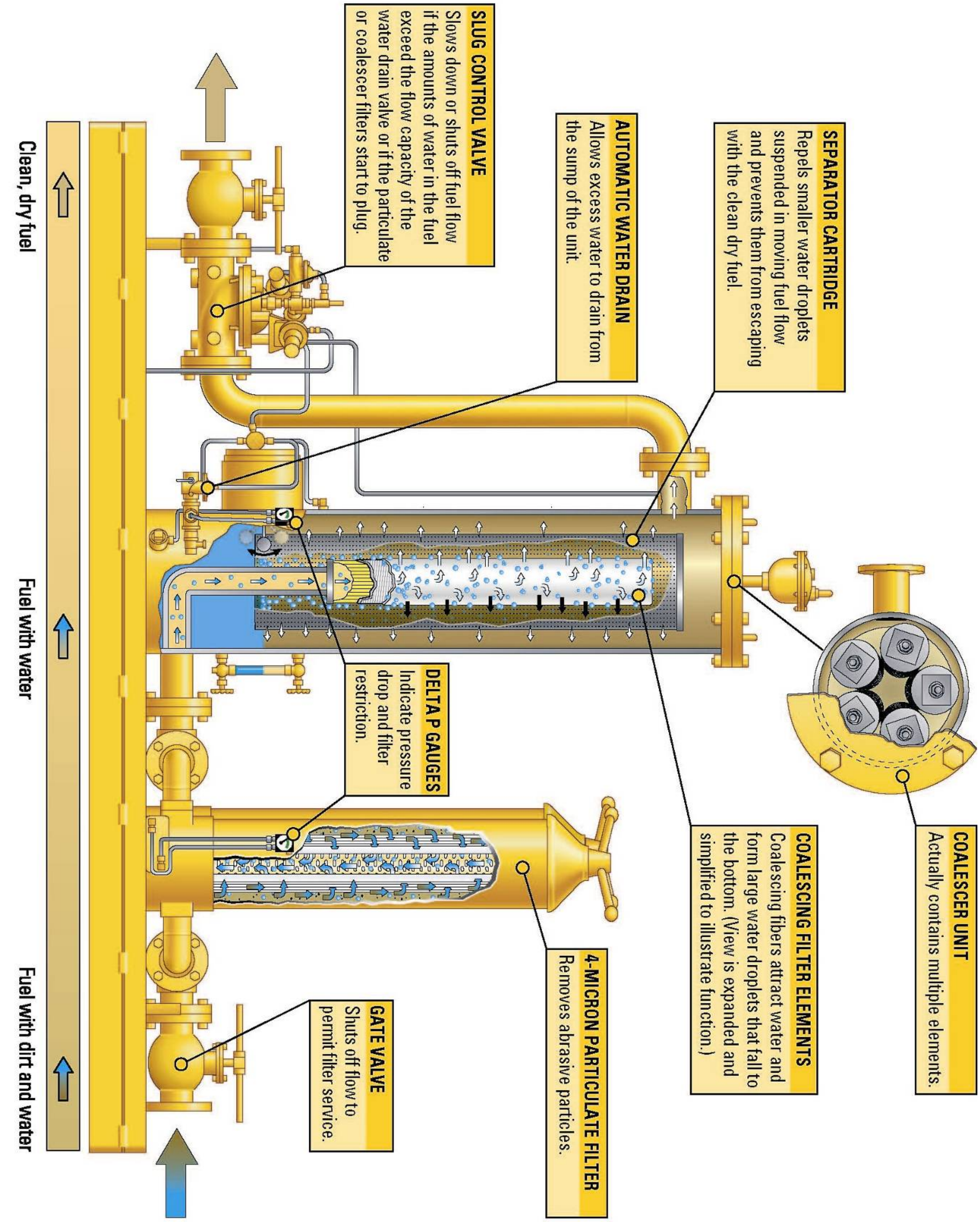
Caterpillar has engineered a packaged system to remove both dirt and water. It requires very little maintenance and contains safeguards to prevent contaminated fuel from passing through the unit. The unit is self-contained on a skid and is located between the fuel storage tank and fueling station. It provides single pass filtration and is offered in four sizes depending on the maximum flow rate of the fuel delivery system.

2.1 Dirt (Particulate Filter)

4-micron, beta 200, full synthetic particulate filter elements remove dirt in a single pass. Filter change intervals of up to one month depending on the level of fuel contamination.

2.2 Water (Coalescing Filter)

Coalescer unit contains multiple elements capable of removing up to 3% water by volume to 1,000 ppm (0.1%) or less at the rated flow. Water is automatically drained, requiring no manual intervention. Coalescing elements do not plug and usually require changing only once a year.



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2.3 Flow Control Valve

An automated flow control valve slows down or stops fuel outlet flow if particulate filters plug or there are massive amounts of water in the fuel. This assures only clean fuel leaves the unit.

3.0 Implementation Steps

Technical information and pricing on these units is available from Cat Global Mining and the Cat Filters and Fluids group. Unit sizing is determined by the maximum flow rate of the fuel delivery system. Four different sizes are available:

50 &100 GPM
Small units intended for remote day tank applications or for portable use on a fuel truck where fueling is done manually.

200 GPM
Intended for fuel stations using fast-fill where maximum flow does not exceed 200 gpm. This unit will handle trucks through 793.

300 GPM
Intended for fast-fill of 797 trucks where fuel flow rates exceed 200 gpm.

4.0 Benefits

Supplying clean fuel to the machine permits the onboard fuel filters to function properly without plugging.

Most injectors should last to engine overhaul and provide improved long-term fuel economy and engine performance.

5.0 Resources Required

Permanent installation requires only a small concrete pad downstream of the bulk storage tank and supply pump. The unit is installed in series in the fuel supply line to the fueling station.

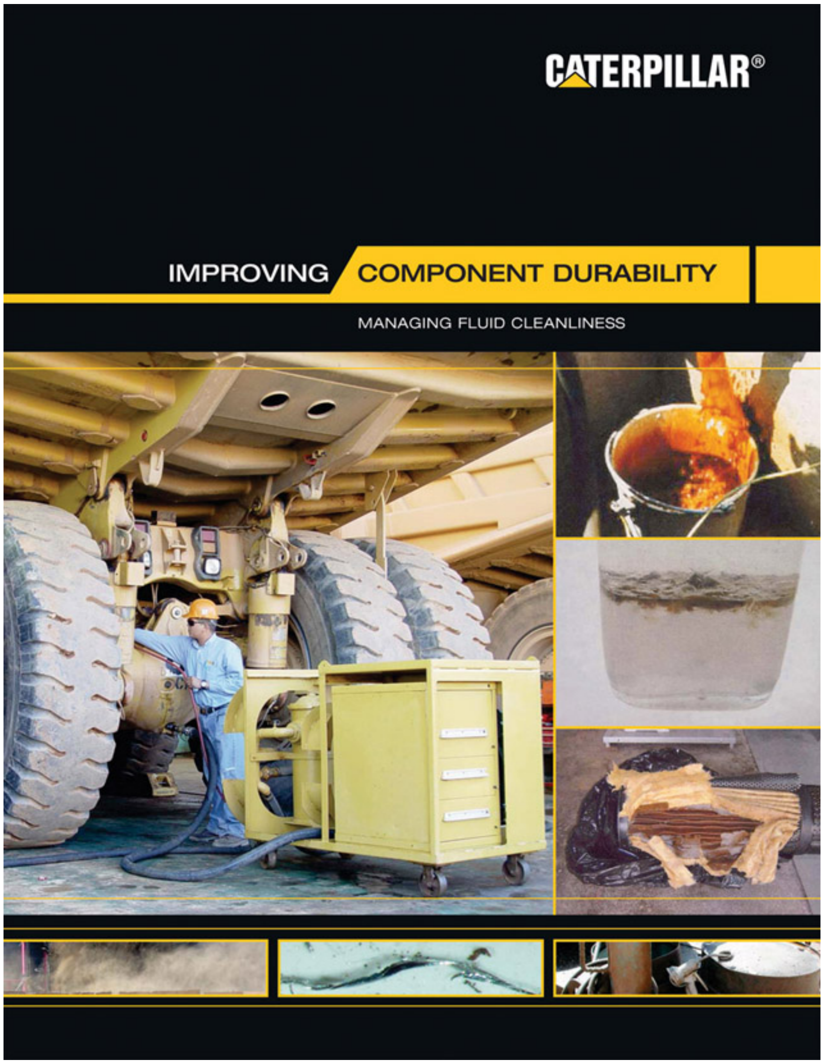
A water container is required nearby to store the waste-water removed from the fuel. No electrical power is required for the unit unless it is used in freezing climates.

An optional electric heating element is available to prevent water from freezing in the bottom of the coalescer tank.

6.0 Supporting Attachments

Component Life Management Master Document, PDF file. (Click on Attachments within this document to view)

An explanation of how the unit works, along with visuals is available in the *"Managing Fluid Cleanliness"* booklet form *SEBF1020*.



7.0 Related Best Practices

0806-2.10-1005	-Bulk Oil Filtration
0806-2.10-1000	-Managing Fluid Cleanliness

8.0 Acknowledgements

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